

Surrogate Substitution Preserves PHI Detectability: A Multi-Detector Equivalence Study

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Abstract

Structure-preserving de-identification replaces protected health information (PHI) with realistic same-type *surrogates*—“Anna S.” becomes “Maria S.”, not [NAME]—so that clinical text stays fluent and downstream tools keep working. But this only helps if the substitution does not itself corrupt the signal those tools rely on. We ask a narrow, testable question: *on the spans a de-identifier actually masks, can downstream PHI detectors still find the surrogate?* We introduce a paired, multi-detector evaluation protocol that (i) scores utility *only on masked spans*, decoupling *coverage* from *utility*; (ii) uses *equivalence testing* (TOST) rather than null-hypothesis significance testing, which is uninformative at our sample size (56k paired spans); and (iii) builds a surrogate-failure typology separating fixable generator defects from intrinsic detector limits. Across 11 detectors, 7 benchmarks, and 7 languages (1,750 documents), recall on masked spans moves from 76.2% to 75.0%—a change our equivalence test shows is *statistically indistinguishable from zero within a ± 2 -point margin* ($p \approx 1 \times 10^{-8}$), with detector ranking preserved. The residual loss is not detectors getting worse at PHI: it concentrates in *malformed and out-of-distribution* surrogates (truncation Chicago $\rightarrow$ Illinois, salience loss Cedars-Sinai $\rightarrow$ Vidant). A redaction floor and an open-source surrogate baseline confirm the effect is a property of well-formed substitution generally, not one tool. We release the evaluation subsets and scoring code so the protocol can audit any structure-preserving transform.

1 Introduction

De-identification of clinical text has two families of output. **Redaction** deletes or tags PHI (John Smith $\rightarrow$ [NAME]), which is safe but destroys the fluency, layout, and distributional properties that downstream models and human readers depend on. **Structure-preserving** de-identification instead substitutes each PHI value with a plausible, same-type *surrogate* (John Smith $\rightarrow$ Maria Lopez), keeping the document readable and machine-parseable [4]. The second family is increasingly attractive: it lets de-identified data flow into analytics, model training, and even second-pass detection without breaking pipelines built for real text.

The promise of structure preservation rests on an unstated assumption: *the surrogate carries the same downstream signal as the value it replaced*. If substitution silently degrades the very features a PHI detector, an NER model, or a clinical parser relies on, then “structure-preserving” is a misnomer—the transform would be qui-

etly laundering PHI into forms tools can no longer see or handle, which is both a utility problem and, for a second-pass safety net, a privacy problem.

This assumption is rarely tested directly, and testing it well is harder than it looks. Three pitfalls recur:

1. **Coverage confounds utility.** A de-identifier that masks 60% of PHI and one that masks 95% cannot be compared on whole-document F1—the score conflates *how much* it masks with *whether what it masks stays usable*. The two must be measured separately.
2. **The large- N significance trap.** With tens of thousands of paired spans, any non-zero difference is “statistically significant” under a standard test, even when operationally meaningless. A p -value here answers the wrong question.
3. **Aggregate error hides mechanism.** A single “−1.2 points” says nothing about *why* the loss happens—boundary jitter, detector weakness, or

a defect in surrogate generation. Only the last is fixable, and only an error typology tells them apart.

We address all three. Our contributions:

- A **paired multi-detector protocol** scoring utility on masked spans only, decoupling coverage from utility, across 11 detectors $\times$ 7 benchmarks $\times$ 7 languages (§4–§5).
- An **equivalence-testing analysis** (TOST) replacing the uninformative significance test with a bounded-effect claim: the change in detectability is provably within ± 2 points of zero (§6).
- A **surrogate-failure typology** attributing the small residual to surrogate-generation quality rather than detectors getting worse at PHI (§7).
- A set of **comparison experiments**—a redaction upper-bound and an open-source surrogate baseline—that make the result reproducible and isolate what is generic to *any* substitution versus specific to one tool (§5, §6).

The framing is deliberately not “a proprietary tool is good.” It is: *here is how to measure whether a structure-preserving transform preserves utility, rigorously*, with one commercial transform as the case study and open baselines for reproducibility.

2 Related Work

2.1 De-identification and surrogate substitution

Clinical de-identification is a long-studied sequence-labeling problem [31], anchored by the i2b2/n2c2 and MEDDOCAN shared tasks [23, 29] and by systems ranging from rule- and dictionary-based pipelines [25] through recurrent and transformer taggers [7, 16, 20] to instruction-tuned LLMs used zero-shot [21]. Most of this literature optimizes *detection*, typically against the HIPAA Safe Harbor identifier set [30]. The downstream question—*what to put in the PHI’s place*—is comparatively under-studied. Surrogate (“hiding in plain sight”) replacement was proposed to keep de-identified notes realistic and to resist re-identification [4]; subsequent work showed poor surrogates can even aid re-identification (the “parrot” attack), underscoring that surrogate *quality* matters [5]. Our work is orthogonal to the detection literature and to any particular surrogate generator: given

that a span is masked, we ask whether the *replacement* remains detectable and well-formed.

[Citation pending] The commercial transform used as our case study implements a surrogate-generation method covered by a Custodian Labs patent [11]. The formal citation will be finalized once the patent issues; we cite it as the source of the transform rather than describing or extending the method.

2.2 Utility-preservation evaluation

Whether privacy transformations preserve downstream utility is central to privacy-preserving NLP. Prior evaluations typically run a single downstream model on original vs. transformed data and compare end-task scores. Two weaknesses recur: (a) a single downstream model cannot separate “the transform is fine” from “this model is robust,” and (b) whole-corpus metrics mix masked and unmasked content. We address (a) with an 11-detector panel spanning rule-based, fine-tuned, and LLM detectors, and (b) by restricting utility measurement to masked spans.

2.3 The large- N trap and equivalence testing

When samples are large, null-hypothesis significance testing rejects the null for negligible effects; the p -value measures precision, not importance—a hazard increasingly flagged in NLP evaluation [3, 9]. The standard remedy is *equivalence testing*—the two one-sided tests (TOST) procedure [2, 28]—which specifies an equivalence margin Δ and tests whether the effect is provably inside $[-\Delta, +\Delta]$; see Lakens [17], Lakens et al. [18] for practical tutorials. TOST is standard in biostatistics but under-used in NLP evaluation, where large paired corpora make the trap acute. We adopt it as the primary inferential tool and report McNemar’s paired test [24] only to demonstrate the trap.

3 Problem Formulation

Structure-preserving de-identification. A transform T maps a document d to d' by replacing each detected PHI value v (of type τ) with a surrogate s of the same type, leaving all other characters unchanged. Gold PHI spans on d are re-projected onto d' by character-level alignment, giving paired spans (v, s) .

Coverage vs. utility. Two quantities must not be conflated. *Coverage* is the fraction of true PHI that T detects and replaces (a property of T ’s detector). *Utility* is, given that a span was replaced, whether a downstream detector still finds the surrogate s as well as it found v (a

property of T ’s generator and of the surrogate’s realism). We report coverage separately and measure utility *only on the masked-span population* $\{(v, s)\}$. This prevents a low-coverage transform from looking good—or a high-coverage one from looking bad—on a metric that is really about substitution quality.

4 Evaluation Protocol

Figure 1 summarizes the protocol. **Paired multi-detector design.** Each document is scored in two conditions—original d and transformed d' —by the *identical* detector suite D ($|D| = 11$). Because the only change between conditions is the substitution, any per-span change in detection is attributable to the substitution, not to the detector or the document.

Metrics. We report span-level P/R/F1 under three matching modes—*exact* (start+end+type), *type* (type+boundary), *overlap* (any character overlap+type)—and *leakage* = $1 - \text{recall}$, the HIPAA-critical quantity. The headline utility metric is *recall retention on masked spans* = $\text{recall}(d')/\text{recall}(d)$ restricted to $\{(v, s)\}$, under overlap matching (so pure boundary jitter from length changes, “Anna S.”→“Maria S.”, is not charged as a miss).

Equivalence testing. For pooled masked-span recall we run TOST with margin $\Delta = 2$ points: we reject non-equivalence iff the 90% CI of the recall difference lies entirely within $[-2, +2]$. We also report McNemar’s test to illustrate the large- N trap. We sweep $\Delta \in \{1, 2, 3\}$.

Error attribution. For the lost population (found on d , missed on d') we hand-code each span into a small failure typology (§7) and check length-preservation to separate boundary artifacts from genuine misses.

5 Experimental Design

Detector panel D . Eleven detectors, three families: *rule/statistical*—Microsoft Presidio [25] (built on spaCy [15]), OBI deid_roberta [16], a RoBERTa tagger [8, 19]; *open LLMs (local)*—Gemma 4 31B / E4B [13], Qwen 3.5-{4B, 9B, 35B-A3B} [27], Llama 3.1-8B / 3.3-70B [14], DeepSeek V2-Lite [6]; *frontier API*—OpenAI GPT-5 [26]. All are Transformer-based [32]. The panel is deliberately heterogeneous: if the equivalence result held only for one architecture it would be a model artifact, not a property of the transform. (Model-version citations point to the closest published technical report for each family. One further model, Moonshot Kimi-VL-A3B, was attempted

but excluded because it required non-standard model-loading code.) LLM detectors emit free-text or JSON identifier lists; we recover character spans by exact-then-fuzzy string search over the source document, so all detectors are scored on the same span basis.

Benchmarks (7; 250 docs each, 1,750 total). ASQ-PHI (English clinical queries), MEDDOCAN (Spanish clinical) [23], MultiCoNER v2 (multilingual NER) [12, 22], and PII-Masking-300k [1] in English, Dutch, French, German. Together: 7 languages, clinical and general-PII text, free-form and structured (JSON) formats. The transform under test is Custodian Guardian Layer `transform` mode (top-1 surrogate, `pii_entities=ALL`).

Comparison conditions. The core study answers “does *this* transform preserve detectability.” Three contrasts isolate *why* and *how generally*:

- **C1—Redaction upper-bound.** Replace each masked value with `*****` (no surrogate signal) and re-detect; the transform—redact gap quantifies what structure preservation buys.
- **C2—Open-source surrogate baseline.** Replace the commercial generator with an open one (Faker [10]) on the *same* masked spans; tests whether the result is generic to substitution or tool-specific, and makes the pipeline reproducible.
- **C3—Per-benchmark equivalence.** Run TOST within each benchmark, sweeping Δ , to check the pooled claim is not an averaging artifact.

The core paired study, the equivalence analysis (pooled and per-benchmark), the redaction floor, the open-surrogate baseline, and the error typology are complete; C1/C2 are currently demonstrated with the CPU detector (Presidio), with a full-panel extension noted as the next strengthening step.

6 Results

Detectability is statistically unchanged. Restricting to the 56,174 masked spans and pooling all 11 detectors, recall moves 76.2% → 75.0% (−1.2 pts; 95% CI $[-1.5, -1.0]$). The TOST equivalence test ($\Delta = 2$) rejects non-equivalence at $p \approx 1 \times 10^{-8}$: the change is provably bounded within ± 2 points of zero. Ranking is preserved; the flagship Llama 3.3-70B is essentially unchanged ($\Delta F1 +0.003$). The same contingency is

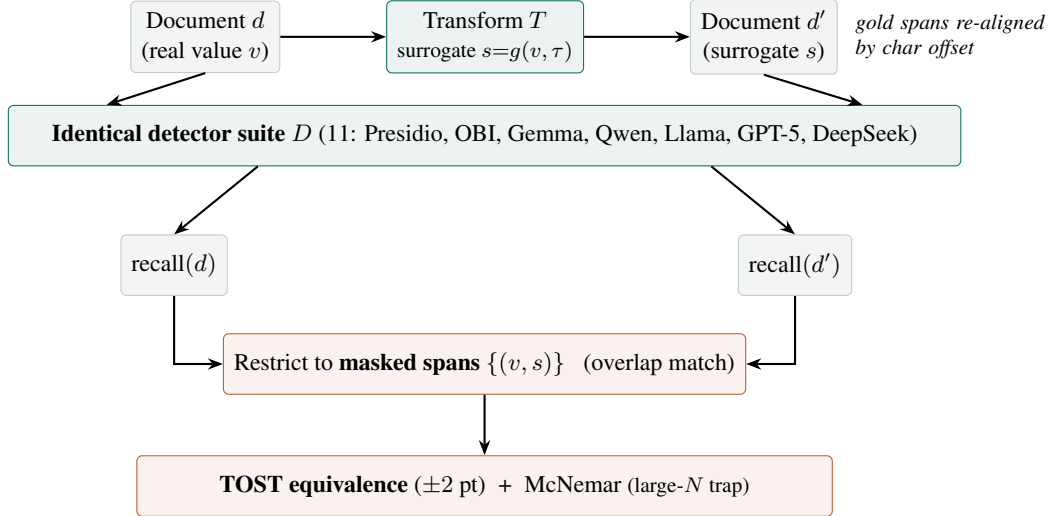


Figure 1: The paired evaluation protocol. Each document is scored in *both* conditions (d , d') by the identical 11-detector suite, so any per-span change is attributable to the substitution. Utility is measured *only on masked spans*, and the paired recall difference is assessed with an equivalence test (TOST) rather than a significance test (§4).

“significant” under McNemar’s test—a direct demonstration of the large- N trap.

Whole-document view (conservative). Before restricting to masked spans, Table 1 reports span-level F1 and leakage on *all* gold spans, per detector. Whole-document Δ F1 ranges from +0.003 (Llama 3.3-70B) to -0.047 (Qwen 3.5-35B-A3B), and leakage rises by only +0.3 to +4.1 points. This view is deliberately pessimistic—it mixes masked spans with exact-boundary penalties on length-changed surrogates and with spans the transform never touched—yet it already bounds the worst case (no detector’s mean F1 moves by more than 4.7 points) and preserves ranking across a $100\times$ span of detector quality (F1 $0.76 \rightarrow 0.04$). The masked-span analysis below isolates the substitution effect from this dilution.

Recall retention on masked spans (overlap). Restricting to masked spans removes that dilution. When the transform masks a span, detectors still find the surrogate 93–100% of the time (Table 2); the ~ 3 -point exact-boundary drop is a length-jitter artifact that vanishes under overlap matching. (The two floor detectors, Llama 3.1-8B and OBI `deid_roberta`, are omitted from Table 2: their original recall is so low that the retention ratio is dominated by noise.)

Per-benchmark equivalence (C3). Table 3 (57,112 masked spans, via `scripts/analyze_equivalence.py`) shows

System	F1		Δ	Leak	
	orig	transf		orig	transf
Gemma 4 31B	.755	.710	−.045	.255	.285
Gemma 4 E4B	.737	.698	−.038	.276	.311
Llama 3.3-70B	.725	.728	+ .003	.245	.262
Qwen 3.5-35B-A3B	.715	.668	−.047	.254	.295
OpenAI GPT-5	.705	.674	−.032	.308	.333
Qwen 3.5-9B	.655	.621	−.034	.391	.422
Qwen 3.5-4B	.567	.533	−.034	.483	.515
Presidio	.416	.398	−.018	.553	.570
DeepSeek V2-Lite	.409	.384	−.025	.672	.696
Llama 3.1-8B	.391	.376	−.015	.633	.650
OBI <code>deid_roberta</code>	.041	.040	−.002	.941	.944

Table 1: Whole-document view (all 11 detectors), mean over 7 benchmarks. Span-level F1 (type match) and leakage (1–recall). Conservative: it includes untouched spans and boundary penalties. Ranking is preserved and Δ F1 is bounded.

the pooled equivalence is *not uniform*: four benchmarks are equivalent within ± 2 points; MEDDOCAN and PII-nl require ± 3 (both dense, identifier-heavy, non-English—where surrogate generation is hardest, §7); MultiCoNER has too few masked spans (220) to test. The honest claim: *detectability is equivalent within ± 2 points on average and on clean text, and within ± 3 on the hardest identifier-dense text*—the residual is

System	Exact	Overlap
Gemma 4 31B	93.1	99.8
Qwen 3.5-9B	92.0	100.0
Qwen 3.5-35B-A3B	90.0	98.4
Llama 3.3-70B	91.4	98.3
GPT-5	91.9	98.3
Gemma 4 E4B	88.9	98.2
Qwen 3.5-4B	84.7	98.0
Presidio	91.3	97.4
DeepSeek V2-Lite	87.7	92.8

Table 2: Recall retention on masked spans (%), transformed÷original.

Benchmark	Masked	O→T	McN. χ^2	Margin
ASQ-PHI	5,427	91.9→91.1	8.0	$\pm 2 \checkmark$
MEDDOCAN	31,978	73.4→71.5	92.4	$\pm 3 \checkmark$
MultiCoNER	220	71.8→71.4	0.0	underpow.
PII en	5,181	76.0→75.1	4.0	$\pm 2 \checkmark$
PII nl	3,757	78.2→76.3	13.1	$\pm 3 \checkmark$
PII fr	5,984	75.2→75.9	3.5	$\pm 2 \checkmark$
PII de	4,565	76.3→76.5	0.4	$\pm 2 \checkmark$
Pooled	57,112	76.1→74.9	82.1	$\pm 2 \checkmark$

Table 3: Per-benchmark equivalence. O→T = masked-span recall, original→transform (%). Pooled TOST $\Delta=2$: $p=3\times 10^{-9}$.

concentrated and attributable, not diffuse degradation.

Redaction floor (C1). Replacing masked values with ***** and re-detecting with Presidio (Table 4): transform recall tracks the *original* within a few points, while redaction collapses to $\approx 0\text{--}4\%$ (the residual is spurious overlap on the * run). Structure-preserving substitution preserves essentially the entire detectability that redaction destroys.

Open-surrogate baseline (C2). Replacing the commercial generator with Faker on the same masked spans (Table 5) yields two findings. *Generality*: an open substitutor preserves masked-span recall at least as well as the original—detectability preservation is a property of well-formed same-type substitution, not one vendor, and reproduces without proprietary access. *The residual is generator quality*: Faker, emitting clean canonical values, *exceeds* the commercial transform by 18–20 points on the hard non-English benchmarks, precisely where the commercial surrogates truncate or garble (§7).

Leakage barely moves (+1.7 to +4.1 pts across

Benchmark	Masked	Orig.	Transf.	Redact
ASQ-PHI	531	94.5	95.7	0.0
MEDDOCAN	2,925	75.4	72.9	0.0
PII en	471	75.6	73.5	1.9
PII nl	373	83.6	77.7	3.8
PII fr	544	61.4	59.9	2.2
PII de	415	56.1	58.3	0.0

Table 4: C1 redaction floor: Presidio masked-span recall (%).

Benchmark	Masked	Orig.	Custodian	Faker
ASQ-PHI	531	94.5	95.7	97.7
MEDDOCAN	2,925	75.4	72.9	79.7
PII en	471	75.6	73.5	89.2
PII nl	373	83.6	77.7	93.0
PII fr	544	61.4	59.9	78.5
PII de	415	56.1	58.3	78.1

Table 5: C2 open-surrogate baseline: Presidio masked-span recall (%). Detector-independent argument; Presidio-only demonstration.

systems), so surrogates are not systematically easier to miss than the PHI they replace.

Where the loss lands (per benchmark). Table 6 breaks whole-document $\Delta F1$ out by benchmark and detector. The pattern is sharp: almost all of the loss concentrates on **ASQ-PHI**—short, sparse-PHI adversarial queries where a single substitution dominates the document score (-0.09 to -0.18)—while the other six benchmarks are essentially flat ($|\Delta F1| \leq 0.06$, most ≤ 0.03). Even the fine-tuned OBI tagger and the rule-based Presidio move by at most a few thousandths on most benchmarks. This is the first sign that the effect tracks *surrogate-generation difficulty* (short adversarial text, dense non-English identifiers) rather than detector family—made precise in the error analysis (§7).

What the transform does (examples). Table 7 shows the substitution qualitatively across languages and formats: each PHI value becomes a same-type surrogate while clinical shorthand, foreign-language syntax, and JSON structure are left byte-for-byte intact. On the well-formed English cases every surrogate is still detected.

Coverage, reported separately. All results above are measured *on the spans the transform masks*. Coverage—how much true PHI it detects and replaces—is a distinct, detector-side axis. It tracks how closely

System	ASQ-PHI	MEDDOCAN	PII en	PII nl	PII fr	PII de	MultiCoNER
Gemma 4 31B	−.175	−.017	−.043	−.038	−.014	−.016	−.009
Gemma 4 E4B	−.106	−.060	−.032	−.021	−.018	−.025	−.007
Qwen 3.5-35B-A3B	−.132	−.034	−.026	−.034	−.040	−.054	−.007
OpenAI GPT-5	−.098	−.046	−.049	−.019	+ .008	−.005	−.014
Qwen 3.5-9B	−.123	−.017	−.025	−.010	−.034	−.024	−.008
Qwen 3.5-4B	−.086	−.055	−.034	−.028	−.015	−.023	+ .002
Presidio	+ .003	−.037	−.032	−.015	−.020	−.015	−.007
Llama 3.1-8B	−.108	+ .023	−.014	+ .007	−.009	+ .003	−.006
OBI deid_roberta	+ .001	−.008	−.004	−.001	+ .003	−.002	−.001

Table 6: Per-benchmark whole-document $\Delta F1$ (transformed − original; negative = drop). Almost all loss lands on ASQ-PHI; the other six benchmarks are flat. Llama 3.3-70B and DeepSeek V2-Lite omitted for space (both flat; see Table 1 for their pooled $\Delta F1$).

Case	Original	Transformed
ASQ-PHI (clinical query)	...a 34-year-old female ...like Anna S., ...at Methodist Hospital on April 12, 2023?	...a 35-year-old female ...like Maria S., ...at Methodist Hospital on March 13, 2021?
ASQ-PHI (short-hand)	Rec mgmt of 70yo M w/ CHF, seen by Dr. John L. at Mt. Sinai on Feb 21, 2023. alt tx options...	Rec mgmt of 73yo M w/ CHF, seen by Dr. James L. at Mt. Egypt on Nov 19, 2021. alt tx options...
PII (German)	...ermächtigen hiermit Monsignore, ...Mit Datum 23/07/2011...	...ermächtigen hiermit Fulgenzio, ...Mit Datum 24/07/2011...
PII (French, JSON)	{ "Date": "20/05/2022", "City": "Saint-Priest", "Username": "phprosdocimo" }	{ "Date": "21/05/2023", "City": "Saint-Priest", "Username": "phprosdocimo" }

Table 7: Worked transform examples. Only the sensitive value moves; abbreviations (w/ CHF, alt tx), non-English syntax, and JSON keys/quotes/indentation are preserved, so downstream parsers and detectors keep working. In the JSON case the transform changes only the date value and leaves the (also-sensitive-looking) username untouched—a coverage decision, not a utility one (see below).

a benchmark’s annotation matches the transform’s notion of sensitive content: it masks **80.9%** of gold PHI on ASQ-PHI and **48.3%** on MEDDOCAN, and less ($\approx 26\%$) on general-domain NER/PII corpora whose annotated entities (encyclopedic names, generic places) fall outside that scope; a configuration sweep confirmed `domain=General` maximizes coverage. A per-entity-type breakdown and a detection-vs-replacement diagnosis (the coverage gap is mostly a *replacement*-step issue— $\approx 77\%$ of missed clinical identifiers were flagged by the transform’s own detector but not substituted) are in Appendix A. Decoupling matters because the two failure modes have different owners and fixes: an under-masked span is a *detection/replacement* miss, whereas a masked-but-missed surrogate is a *generation* defect

(§7). Conflating them on whole-document F1 (Table 1) would let a high-coverage/low-quality transform and a low-coverage/high-quality one look identical. We therefore report coverage as context and reserve the utility claim for masked spans.

Matching modes. Unless noted, retention uses *overlap* matching (detector flags any part of the surrogate span). Exact matching (Table 2, left) additionally requires identical character boundaries and is therefore sensitive to surrogate length changes (“Anna S.”→“Maria S.” shifts the end offset); the ~ 3 -point exact–overlap gap is this boundary jitter, not missed PHI. Type matching (boundary + type) sits between the two and tracks overlap closely.

7 Error Analysis

The lost population. Pooled across detectors and masked spans: 39,550 spans found in both conditions; 3,261 lost. Critically, 51% of lost spans are the *same length* as the original—not a boundary effect. By type: LOCATION 28%, NAME 23%, DATE/AGE 22%, ID/contact 20%.

Three failure modes—all generator-side.

(1) *Malformed/truncated surrogates* (largest cause): Chicago → **Illino**, El Paso → **El**, Ciudad de la Habana → **Cuidad de la Havana**. The fragment no longer matches the lexical pattern detectors learned for real names/places; this is why span-level loss is highest on MEDDOCAN (6.9%; Spanish, dense, identifier-heavy) and lowest on clean English ASQ-PHI (2.6%). (2) *Loss of salience*: a canonical entity replaced by an obscure one (Cedars-Sinai → **Vidant**); detectors partly rely on pre-training familiarity, so swapping a famous value for a rare one removes the prior. Inherent to any value substitution; mainly costs weaker detectors. (3) *x-masking of IDs/emails*: nachorutor@... → **nxxxxxxxxx@...**. The x run preserves format but breaks the realistic-token pattern. This case is *privacy-positive*—the original value is destroyed—even though it counts against recall.

Implication. Detectors are not getting worse at PHI; the small recall gap is driven by surrogate-generation quality (truncation, garbling, salience, x-masking). C2 confirms this directly: an open generator emitting clean canonical values *exceeds* the commercial transform by 18–20 points on exactly the hard non-English benchmarks where its surrogates truncate—so the residual tracks generator quality, not the act of substitution.

8 Discussion

What the evidence supports. Structure-preserving substitution does not hide well-formed PHI from downstream detection. Because the effect is equivalence-bounded within ± 2 points and ranking-preserving across a heterogeneous 11-detector panel, a transform of this quality can be safely inserted ahead of detection/analytics pipelines built for real clinical text. **Generality (C2).** Faker preserves masked-span recall at least as well as the original across all six benchmarks, so ± 2 points is a property of well-formed substitution, not one vendor; where the commercial generator trails, a cleaner generator closes the gap, locating the residual squarely in generation quality. **A reusable protocol.** The paired

masked-span design and the TOST margin are not specific to one vendor or language—a template for auditing any structure-preserving privacy transform.

9 Ethics and Data Statement

All benchmarks are public or synthetic (ASQ-PHI synthetic; MEDDOCAN released for a shared task; PII-Masking-300k synthetic; MultiCoNER v2 public). **No real patient data is used.** We release the 250-document subsets and scoring code with license notes. **Dual-use.** A utility-preserving de-identifier could in principle launder identifiable data into a fluent form; our masked-span/leakage reporting and the privacy-positive framing of x-masking keep the privacy accounting explicit. **Conflict of interest.** The commercial transform (Custodian Guardian Layer) is developed by Custodian AI, with which the authors are affiliated; this work was conducted with Custodian AI’s support. To limit bias we (i) frame the contribution as a reusable protocol, (ii) include open-source baselines so results are reproducible without proprietary access, and (iii) report leakage and coverage alongside utility.

10 Limitations

Coverage is reported but not the focus; a transform can preserve utility on what it masks while under-masking (the axes are independent by design). 250 docs/benchmark bounds per-benchmark power (though the pooled masked-span N is large). C1/C2 are currently demonstrated with a CPU detector (Presidio); a full 11-detector extension is future work. LLM detectors are prompt-sensitive; we fix one prompt per model and release it with the code and data at the project’s reproducibility page. Finally, all seven benchmarks are public or synthetic; validating the protocol on real-EHR corpora (n2c2, MIMIC-IV-Note, CARMEN-I) is future work pending the relevant data-use agreements.

A Coverage Details

Coverage (§6) is the fraction of a benchmark’s gold PHI whose characters the transform actually changed (diffib alignment). Table 8 breaks it down by entity type on the clinical sets (ASQ-PHI + MEDDOCAN). High-frequency free-text types (names, dates) are masked well; the weak spots are high-sensitivity structured identifiers and geography—exactly the HIPAA Safe Harbor items that most need masking.

Detection vs. replacement. On a sample of

Entity type (clinical)	Coverage
NAME / person	75.0%
DATE / age	73.4%
ID / contact (MRN, patient ID, phone, email)	50.7%
ORG / facility	46.6%
LOCATION / address	43.7%

Table 8: Coverage by entity type on clinical benchmarks. Overall the transform altered 39% of annotated PII (54% on clinical sets); general-domain corpora sit near 26%.

8 MEDDOCAN documents (62 missed ID/location spans), 48/62 ($\approx 77\%$) of the *unmasked* identifiers were nonetheless flagged as sensitive by the transform’s own detector—they were detected but not substituted; only 14/62 (23%) were undetected. The coverage gap is therefore mostly a *replacement-step* issue (act on everything the detector surfaces), which is more tractable than raising detection recall. (Small sample; “detected” judged by loose token overlap, so 77% is directional.) Concrete unmasked HIPAA identifiers included patient IDs (80926, 7845693), care-contact IDs (4387684), and facilities/geography (Hospital de Cruces, España, postal codes 41005, 28047).

B Representative Per-Detector Examples

All-detector overlay on one document (“ranking proof”). All eleven detectors’ predictions on one transformed document (asq_00001), overlaid on the text and sorted by this document’s F1: **green** = correctly caught PHI, **yellow** = false positive, **red** = missed PHI (leakage). The four gold surrogate values are John T., the x-masked location Saint. Vxxxxxxx, April 25th, 2018, and the ID 987654321. Eight of eleven detectors tag all four cleanly (F1 100); the two floor detectors, Presidio and DeepSeek, miss the x-masked location surrogate (§7 mode 3)—DeepSeek also drops the numeric ID—and OBI over-fragments boundaries and false-fires on the age 60. The surrogate values shown are the transform’s output, not the original PHI.

Gold (reference)	4 PHI spans
Evaluation of long-term outcomes for bypass surgery in patients over 60, referencing Mr. John T., operated at Saint. Vxxxxxxx on April 25th, 2018 (ID: 987654321)?	

OpenAI GPT-5	tp 4 · fp 0 · missed 0 · F1 100%
Evaluation of long-term outcomes for bypass surgery in patients over 60, referencing Mr. John T., operated at Saint. Vxxxxxxx on April 25th, 2018 (ID: 987654321)?	
Llama 3.1-8B	tp 4 · fp 0 · missed 0 · F1 100%
Evaluation of long-term outcomes for bypass surgery in patients over 60, referencing Mr. John T., operated at Saint. Vxxxxxxx on April 25th, 2018 (ID: 987654321)?	
Qwen 3.5-4B	tp 4 · fp 0 · missed 0 · F1 100%
Evaluation of long-term outcomes for bypass surgery in patients over 60, referencing Mr. John T., operated at Saint. Vxxxxxxx on April 25th, 2018 (ID: 987654321)?	
Gemma 4 31B	tp 4 · fp 0 · missed 0 · F1 100%
Evaluation of long-term outcomes for bypass surgery in patients over 60, referencing Mr. John T., operated at Saint. Vxxxxxxx on April 25th, 2018 (ID: 987654321)?	
Llama 3.3-70B	tp 4 · fp 0 · missed 0 · F1 100%
Evaluation of long-term outcomes for bypass surgery in patients over 60, referencing Mr. John T., operated at Saint. Vxxxxxxx on April 25th, 2018 (ID: 987654321)?	
Qwen 3.5-35B-A3B	tp 4 · fp 0 · missed 0 · F1 100%
Evaluation of long-term outcomes for bypass surgery in patients over 60, referencing Mr. John T., operated at Saint. Vxxxxxxx on April 25th, 2018 (ID: 987654321)?	
Gemma 4 E4B	tp 4 · fp 0 · missed 0 · F1 100%
Evaluation of long-term outcomes for bypass surgery in patients over 60, referencing Mr. John T., operated at Saint. Vxxxxxxx on April 25th, 2018 (ID: 987654321)?	
Qwen 3.5-9B	tp 4 · fp 0 · missed 0 · F1 100%
Evaluation of long-term outcomes for bypass surgery in patients over 60, referencing Mr. John T., operated at Saint. Vxxxxxxx on April 25th, 2018 (ID: 987654321)?	
OBI deid_roberta	tp 4 · fp 1 · missed 0 · F1 89%
Evaluation of long-term outcomes for bypass surgery in patients over 60, referencing Mr. John T., operated at Saint. Vxxxxxxx on April 25th, 2018 (ID: 987654321)?	
Microsoft Presidio	tp 3 · fp 1 · missed 1 · F1 75%
Evaluation of long-term outcomes for bypass surgery in patients over 60, referencing Mr. John T., operated at Saint. Vxxxxxxx on April 25th, 2018 (ID: 987654321)?	
DeepSeek V2-Lite	tp 2 · fp 0 · missed 2 · F1 67%
Evaluation of long-term outcomes for bypass surgery in patients over 60, referencing Mr. John T., operated at Saint. Vxxxxxxx on April 25th, 2018 (ID: 987654321)?	

A harder case: a dense Spanish clinical header. Figure-style overlay on a 260-character window of a MEDDOCAN document (S0210-...008-1), where the same eleven detectors diverge far more—F1 from 96

down to 0. On dense, identifier-heavy non-English text the *per-document* ranking reshuffles (here DeepSeek and Qwen-4B lead, while Llama 3.1-8B finds nothing in this window), even though the *aggregate* ranking over 250 documents is stable (Table 1). The garbled surrogate Ciudad Real (from Ciudad Real) is still caught by most detectors; the misses cluster on the professional-licence ID 03 14 16485, the sex fields, and the repeated age—the identifier types with the weakest coverage and the hardest surrogates (Appendix A).

Gold (reference)	12 PHI spans
Ciudad Real. CP: 13002. Datos asistenciales. Referencia de nacimiento: 24/10/1963. País: España. Mejores: 53 años Sexo: H. Referencia de Ingreso: 14/12/2017. Episodio: 746589123. Médico: Luis Ruiz Camuñas N°Col: 03 14 16485. Historia Actual: Varón de 53 años q	
DeepSeek V2-Lite	tp 11 · fp 0 · missed 1 · F1 96%
Ciudad Real. CP: 13002. Datos asistenciales. Referencia de nacimiento: 24/10/1963. País: España. Mejores: 53 años Sexo: H. Referencia de Ingreso: 14/12/2017. Episodio: 746589123. Médico: Luis Ruiz Camuñas N°Col: 03 14 16485. Historia Actual: Varón de 53 años q	
Qwen 3.5-4B	tp 11 · fp 0 · missed 1 · F1 96%
Ciudad Real. CP: 13002. Datos asistenciales. Referencia de nacimiento: 24/10/1963. País: España. Mejores: 53 años Sexo: H. Referencia de Ingreso: 14/12/2017. Episodio: 746589123. Médico: Luis Ruiz Camuñas N°Col: 03 14 16485. Historia Actual: Varón de 53 años q	
OBI deid_roberta	tp 10 · fp 0 · missed 2 · F1 91%
Ciudad Real. CP: 13002. Datos asistenciales. Referencia de nacimiento: 24/10/1963. País: España. Mejores: 53 años Sexo: H. Referencia de Ingreso: 14/12/2017. Episodio: 746589123. Médico: Luis Ruiz Camuñas N°Col: 03 14 16485. Historia Actual: Varón de 53 años q	
Qwen 3.5-9B	tp 10 · fp 0 · missed 2 · F1 91%
Ciudad Real. CP: 13002. Datos asistenciales. Referencia de nacimiento: 24/10/1963. País: España. Mejores: 53 años Sexo: H. Referencia de Ingreso: 14/12/2017. Episodio: 746589123. Médico: Luis Ruiz Camuñas N°Col: 03 14 16485. Historia Actual: Varón de 53 años q	
Gemma 4 31B	tp 9 · fp 0 · missed 3 · F1 86%
Ciudad Real. CP: 13002. Datos asistenciales. Referencia de nacimiento: 24/10/1963. País: España. Mejores: 53 años Sexo: H. Referencia de Ingreso: 14/12/2017. Episodio: 746589123. Médico: Luis Ruiz Camuñas N°Col: 03 14 16485. Historia Actual: Varón de 53 años q	

Qwen 3.5-35B-A3B	tp 9 · fp 1 · missed 3 · F1 82%
Ciudad Real. CP: 13002. Datos asistenciales. Referencia de nacimiento: 24/10/1963. País: España. Mejores: 53 años Sexo: H. Referencia de Ingreso: 14/12/2017. Episodio: 746589123. Médico: Luis Ruiz Camuñas N°Col: 03 14 16485. Historia Actual: Varón de 53 años q	
OpenAI GPT-5	tp 8 · fp 0 · missed 4 · F1 80%
Ciudad Real. CP: 13002. Datos asistenciales. Referencia de nacimiento: 24/10/1963. País: España. Mejores: 53 años Sexo: H. Referencia de Ingreso: 14/12/2017. Episodio: 746589123. Médico: Luis Ruiz Camuñas N°Col: 03 14 16485. Historia Actual: Varón de 53 años q	
Llama 3.3-70B	tp 8 · fp 0 · missed 4 · F1 80%
Ciudad Real. CP: 13002. Datos asistenciales. Referencia de nacimiento: 24/10/1963. País: España. Mejores: 53 años Sexo: H. Referencia de Ingreso: 14/12/2017. Episodio: 746589123. Médico: Luis Ruiz Camuñas N°Col: 03 14 16485. Historia Actual: Varón de 53 años q	
Gemma 4 E4B	tp 8 · fp 0 · missed 4 · F1 80%
Ciudad Real. CP: 13002. Datos asistenciales. Referencia de nacimiento: 24/10/1963. País: España. Mejores: 53 años Sexo: H. Referencia de Ingreso: 14/12/2017. Episodio: 746589123. Médico: Luis Ruiz Camuñas N°Col: 03 14 16485. Historia Actual: Varón de 53 años q	
Microsoft Presidio	tp 6 · fp 1 · missed 6 · F1 63%
Ciudad Real. CP: 13002. Datos asistenciales. Referencia de nacimiento: 24/10/1963. País: España. Mejores: 53 años Sexo: H. Referencia de Ingreso: 14/12/2017. Episodio: 746589123. Médico: Luis Ruiz Camuñas N°Col: 03 14 16485. Historia Actual: Varón de 53 años q	
Llama 3.1-8B	tp 0 · fp 0 · missed 12 · F1 0%
Ciudad Real. CP: 13002. Datos asistenciales. Referencia de nacimiento: 24/10/1963. País: España. Mejores: 53 años Sexo: H. Referencia de Ingreso: 14/12/2017. Episodio: 746589123. Médico: Luis Ruiz Camuñas N°Col: 03 14 16485. Historia Actual: Varón de 53 años q	

Agreement on well-formed surrogates. On clean English (ASQ-PHI), a plausible same-type surrogate is caught by nearly the whole panel (Table 9): the substitution is invisible to detection. Divergence is confined to the two floor detectors (Llama 3.1-8B, DeepSeek V2-Lite).

Representative losses, by failure type. Table 10 shows genuine “lost” cases (MEDDOCAN): the original value was found by most detectors, but a defective surrogate is found by few. Each maps to one of the three failure types in §7, and the drop is shared across the panel—i.e. it is a property of the surrogate, not of any

Document	Surrogate span (type)	Caught
asq_00000	Maria S. (NAME)	10/11
asq_00000	Methodist Hospital (LOC)	9/11
asq_00000	March 13, 2021 (DATE)	9/11
asq_00003	James L. (NAME)	11/11
asq_00003	Mt. Egypt (LOC)	11/11
asq_00003	Nov 19, 2021 (DATE)	11/11

Table 9: Number of detectors (of 11) that find each well-formed surrogate. Same-type swaps stay broadly detectable.

one detector.

Original → surrogate	O → T	Type
Cuba → Havana	9 → 2	salience
San Fernando → Francsico Luis	8 → 3	garbled
Cádiz → Cadiz	8 → 4	accent stripped
Cecilio	8 → 4	truncation
Pujazón → Pujazón		
ignaciotorne@... → ixxxxxxxxxxx@...	9 → 2	x-mask
23/07/1948 → 2xxxxxxxx	11 → 4	x-mask
40 → 35 (age)	9 → 2	bare number

Table 10: Representative lost spans on MEDDOCAN. O → T = detectors finding the original vs. the surrogate (panel of 11–12; original includes one extra decoding variant). Losses are shared across detectors and align with the §7 typology.

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